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Uncertainty principles for the short-time linear canonical transform of complex signals

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ABSTRACT

The short-time linear canonical transform (STLCT) is a novel time-frequency analysis tool. In this paper, we generalize some different uncertainty principles for the STLCT of complex signals. Firstly, a new uncertainty principle for STLCT of complex signals in time and frequency domains is explored. Secondly, an uncertainty principle in two STLCT domains is obtained. They show that the lower bounds are related to the covariance of time and frequency and can be achieved by complex chirp signals with Gaussian signals. Then the uncertainty principle for the two conditional standard deviations of the spectrogram associated with the STLCT is derived. In addition, an example is also carried out to verify the correctness of the theoretical analyses. Finally, some potential applications are presented to show the effectiveness of the theorems.

1. Introduction

The linear canonical transform (LCT) [1–4] is an integral transform with four parameters (a, b, c, d) and it has many special cases such as the Fourier transform (FT) [5], the fractional Fourier transform (FRFT) [6,7], the Lorentz transform [8], and the Fresnel transform [9]. Comparing to the FRFT with one extra degree of freedom and FT without a parameter, three free parameters of the LCT make it more flexible. The LCT has become a powerful analyzing tool and it has been widely used in some fields, such as applied mathematics, signal processing, optics, quantum physics and filter design [10–16].

Serving as a powerful analyzing tool, the LCT converts a signal from the space domain to the corresponding spectral domain. However, it cannot support the knowledge of the signal at the local position at the same time due to its global kernel. To meet these requests, Kou and Xu [10] first proposed the short-time linear canonical transform (STLCT). The STLCT is also known as the windowed linear canonical transform and modifies the traditional the LCT by utilizing a window. It offers local contents, enjoys high resolution, and eliminates cross terms. In the STLCT, the original signal is multiplied by a window function before performing the LCT, and this window is scanned along the space axis, thus the analysis of the signal in both the space and frequency domains is realized simultaneously. The basic idea of the STLCT is: break up the signal into small time segments and linear canonical analyze each time segment to ascertain the frequencies that existed in that segment. The totality of such spectra indicates how the spectrum is varying in time. The STLCT, which maps the time domain signal into the joint time and frequency domain, has recently attracted some attention in the area of signal processing. The Poisson summation formula and series expansions for the STLCT are derived in [10]. Bahri and Ashino [17] discussed some properties of STLCT and the logarithmic uncertainty principle for the STLCT was established. Zhang [1] obtained a sampling theorem for the STLCT by means of a generalized Zak transform associated with the LCT. Zhang [18] discussed the discrete windowed linear canonical transform. In [19], a dual window solution of the STLCT was investigated. The Lieb's uncertainty principle for the STLCT has been discussed in [20], which takes the LCT version as one of its special cases. In our previous work [21], we proposed the quaternion windowed linear canonical transform and investigated some properties of the quaternion windowed linear canonical transform.

On the other hand, the uncertainty principle plays an important role in signal processing and physics [11,22–29]. In signal processing an uncertainty principle states that the product of the variances of the signal in the time and frequency domains has a lower bound. In quantum mechanics an uncertainty principle asserts that one cannot make certain of the position and velocity of an electron (or any particle) at the same time. That is, increasing the knowledge of the position decreases the knowledge

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of the velocity or momentum of an electron. As we all know, the Heisenberg's uncertainty principles are closely related to the signal moments. Based on the signal moments, Feng, Li and Rassias [30] investigated the weighted Heisenberg-Pauli-Weyl uncertainty principles for the LCT. Zhang [31] derived a new LCT uncertainty principle for complex signals using the absolute spread matrix rotation orthogonal decomposition. In [32], an uncertainty relation in the LCT domain was obtained. It shows that nonzero signal's energy in two arbitrary LCT domains cannot be arbitrarily large simultaneously, which is the generalization of the uncertainty principle for signal concentrations in the Fourier domain. Dang, Deng and Qian [33] obtained the uncertainty principles under the LCT of a complex signal, and a lower bound for the uncertainty product of a signal in two LCT domains was proposed that was sharper than those in the existing literature. In [34], Xu et al. studied the uncertainty relation for the STLCT in two STLCT domains. Huang et al. [35] discussed the uncertainty principle and orthogonal condition for the STLCT in one STLCT domains. Unfortunately, these proofs and results only hold for the real signals. However, the uncertainty principles for the STLCT of complex signals have not been found in the literature. It would be interesting to know if those results could be extended to the STLCT of complex signals in this article.

The difficulties of obtaining these results are as follows: the first one is how to obtain the time and frequency spread of the STLCT that can be used in subsequent the uncertainty principle of the STLCT computation; second challenge is how to extend the uncertainty principle of the LCT to the STLCT; the third challenge is how to obtain the uncertainty principle for the two conditional standard deviations of the spectrogram.

The purpose of this paper is to obtain three kinds of uncertainty principle associated with the STLCT and discuss their potential applications. It is found that the uncertainty principles of STLCT are related to the matrixes parameter, signal and the window function. In the face of these difficulties, we first define the time and frequency spread of the STLCT based on the previous work [25,36]. Then we shall apply the uncertainty principle for the LCT to obtain uncertainty principle associated with the STLCT in time and frequency domains. Furthermore, the uncertainty principle in two STLCT domains is obtained. The derived bounds are related with the phase and the amplitude of the complex signal. They are shown here that these lower bounds can be achieved by Gaussian signals. The uncertainty principle for the two conditional standard deviations associated with the STLCT is discussed by utilizing the Hermitian operators. The advantage of the achieved results is that the phenomenon of different lower bounds for real and complex signals existing in the STLCT domain can be explained clearly with these lower bounds because real signals have zero covariance. Finally, an example is also carried out to verify the correctness of the theoretical analyses, and some potential applications are presented to show the effectiveness of the theorems.

The paper is organized as follows: In Section 2, we present some general definitions. Some different uncertainty principles associated with the STLCT are provided in Section 3. In Section 4, an example and potential applications are given. Some conclusions are drawn in Section 5.

2. Preliminary

In this section we review some relevant contents.

Let $f(t)=x(t)e^{i\phi(t)}\in L^2(\mathbb{R})$ and window function $0\neq g(t)=y(t)e^{i\varphi(t)}\in L^2(\mathbb{R})$, where x and ϕ are the amplitude and phase functions of f; y and φ are the amplitude and phase functions of g. Unless otherwise stated, we assume that the following signals are all complex signals in this paper.

Definition 1. (LCT) Let $A=(a,b,c,d)$ be a matrix parameter satisfying $a,b,c,d\in\mathbb{R}$ and $ad-bc=1$. For a signal $f(t)$, the LCT of $f(t)$ is defined as [37,38]

$$L_A^f(u)=L_A[f(t)](u)=\begin{cases}\int_{-\infty}^{+\infty} f(t)K_A(t,u)\mathrm{d}t, & b\neq 0\\ \sqrt{d}e^{i\frac{cd}{2}u^2}f(du)), & b=0\end{cases} \tag{1}$$

where $t,u\in\mathbb{R}$ and the transform kernel of LCT is given by

$$K_A(t,u)=\frac{1}{\sqrt{i2\pi b}}e^{i\frac{a}{2b}t^2-i\frac{1}{b}tu+i\frac{d}{2b}u^2}. \tag{2}$$

The following properties [39] will be useful in this paper

$$K_A^*(t,u)=K_{A^{-1}}(u,t), \tag{3}$$

$$\int_{-\infty}^{+\infty} e^{\pm iux}\mathrm{d}u=2\pi\delta(x);\quad \int_{-\infty}^{+\infty}\delta(t-u)f(t)\mathrm{d}t=f(u), \tag{4}$$

$$\|f\|^2_{L^2(\mathbb{R})}=\|f\|^2=\int_{-\infty}^{+\infty}|f(t)|^2\mathrm{d}t, \tag{5}$$

where $A^{-1}=(d,-b,-c,a)$. In the following, we only consider $b\neq 0$, since the LCT with $b=0$ is a kind of scaling and chirp multiplication operations [40]. The inverse LCT expression is [38,39]

$$f(t)=\int_{-\infty}^{+\infty} L_A^f(u)K_{A^{-1}}(u,t)\mathrm{d}u. \tag{6}$$

The LCT is a good tool to analyze signals in the frequency domain, but it cannot support the knowledge of the signal at the local position at the same time due to its global kernel. To tackle this problem, by adding a suitable window to the LCT, it is defined as follows [10].

Definition 2. (STLCT) Let g be a window function. $A=(a,b,c,d)$ be a matrix parameter satisfying $a,b,c,d\in\mathbb{R}$ and $ad-bc=1$. The STLCT of a signal f with respect to g is defined by

$$\begin{aligned}S_g^A f(t,u)&=\int_{-\infty}^{+\infty} f(\tau)g^*(\tau-t)K_A(\tau,u)\mathrm{d}\tau\\ &=\int_{-\infty}^{+\infty} f_t(\tau)K_A(\tau,u)\mathrm{d}\tau,\end{aligned} \tag{7}$$

where $f_t(\tau)=f(\tau)g^*(\tau-t)=x(\tau)y^*(\tau-t)e^{i(\phi(\tau)-\varphi(\tau-t))}$, u is a selected analyzing frequency, t is an arbitrary time shift, and $K_A(\tau,u)$ is given by Eq. (2).

By the Definition 2, the result of the STLCT implies information about time t and frequency u. So we can know not only the STLCT frequency domain contents but also how they change with time.

$f_t(\tau)$ is called the local signal, it is the signal we see at time t. A different signal can be seen at a different time.

Let

$$Q(t)=\int_{-\infty}^{+\infty}|f(\tau)g^*(\tau-t)|^2\mathrm{d}\tau, \tag{8}$$

then the local normalized signal is

$$q_t(\tau)=\frac{1}{\sqrt{Q(t)}}f(\tau)g^*(\tau-t). \tag{9}$$

Hence $\int_{-\infty}^{+\infty}|q_t(\tau)|^2\mathrm{d}\tau=1$.

Inspired by Leon Cohen [25], we can define the local spectrum by

$$L_u(\mu)=L_A^f(\mu)L_A^g(u-\mu), \tag{10}$$

where $L_A^f(\mu), L_A^g(\mu)$ are given by Definition 1, μ is the running frequency and u is the fixed frequency of interest. In order to study the time properties of a signal at a particular frequency, we can window the spectrum of the signal, $L_A^f(\mu)$, with a frequency window function, $L_A^g(\mu)$.

Let

$$P(u)=\int_{-\infty}^{+\infty}|L_A^f(\mu)L_A^g(u-\mu)|^2\mathrm{d}\mu, \tag{11}$$

then the normalized local spectrum is

$$p_u(\mu)=\frac{1}{\sqrt{P(u)}}L_A^f(\mu)L_A^g(u-\mu). \tag{12}$$

The short-frequency time transform is

$$s_g^A f(t,u)=\int_{-\infty}^{+\infty}L_A^f(\mu)L_A^g(u-\mu)K_A^*(t,\mu)\mathrm{d}\mu. \tag{13}$$

Hence

$$|S_g^A f(t,u)|^2=|s_g^A f(t,u)|^2. \tag{14}$$

Some series of important properties of the STLCT are listed in [10].

Now, we give a lemma, it is very important in this paper.

Lemma 1. *Let $g\in L^2(\mathbb{R})$ be a window function. For a complex signal $f\in L^2(\mathbb{R})$, we have*

$$S_g^A f(t,u)=\int_{-\infty}^{+\infty}L_A^f(\xi)G^*(\xi|u,t)\mathrm{d}\xi, \tag{15}$$

where $G^*(\xi|u,t)=\sqrt{-i2\pi b}e^{i\frac{d'}{2b}(\xi-u)^2}L_{A_1}^g(\xi-u)^*K_A(t,u)K_A^*(t,\xi)$, $L_{A_1}^g(\xi-u)$ is the LCT of the window function $g(\tau)$, $A_1=(0,b',-\frac{1}{b'},d')$, where $0\neq b'=b\in\mathbb{R}$, the parameter b in the matrix $A=(a,b,c,d)$ and the parameter $d'\in\mathbb{R}$.

Proof. By Definition 2 and Eq. (6), we have

$$\begin{aligned}S_g^A f(t,u)&=\int_{-\infty}^{+\infty}f(\tau)\overline{g(\tau-t)}K_A(\tau,u)\mathrm{d}\tau\\&=\int_{-\infty}^{+\infty}L_A^f(\xi)\int_{-\infty}^{+\infty}K_{A^{-1}}(\xi,\tau)\overline{g(\tau-t)}K_A(\tau,u)\mathrm{d}\tau\mathrm{d}\xi.\end{aligned} \tag{16}$$

Let $G^*(\xi|u,t)=\int_{-\infty}^{+\infty}K_{A^{-1}}(\xi,\tau)\overline{g(\tau-t)}K_A(\tau,u)\mathrm{d}\tau$ and $\tau-t=\eta$, then

$$\begin{aligned}&G^*(\xi|u,t)\\&=\int_{-\infty}^{+\infty}K_{A^{-1}}(\xi,\tau)\overline{g(\tau-t)}K_A(\tau,u)\mathrm{d}\tau\\&=\int_{-\infty}^{+\infty}\overline{g(\eta)}\frac{1}{\sqrt{-i2\pi b}}\frac{1}{\sqrt{i2\pi b}}e^{-i\frac{(u-\xi)}{b}(\eta+t)+i\frac{d}{2b}(u^2-\xi^2)}\mathrm{d}\eta\\&=\frac{1}{\sqrt{i2\pi b}}\int_{-\infty}^{+\infty}\frac{1}{\sqrt{-i2\pi b}}\overline{g(\eta)}e^{i\frac{0}{-2b}\eta^2-i\frac{(\xi-u)}{-b}\eta+i\frac{d'}{-2b}(\xi-u)^2}\mathrm{d}\eta\\&\times e^{i\frac{d'}{2b}(\xi-u)^2+i\frac{d}{2b}(u^2-\xi^2)-i\frac{(u-\xi)}{b}t}\\&=\frac{1}{\sqrt{i2\pi b}}e^{i\frac{d'}{2b}(\xi-u)^2}L_{A_1}^g(\xi-u)^*e^{-i\frac{ut}{b}+i\frac{d}{2b}u^2}e^{i\frac{\xi t}{b}+i\frac{d}{-2b}\xi^2}\\&=\sqrt{-i2\pi b}e^{i\frac{d'}{2b}(\xi-u)^2}L_{A_1}^g(\xi-u)^*K_A(t,u)K_A^*(t,\xi).\quad\square\end{aligned} \tag{17}$$

3. Uncertainty principles for the STLCT

Time-frequency distributions providing high resolution have been desirable [41,42]. The STLCT [10,17,20] is a useful tool of spatial-frequency analysis, which is able to analyze non-stationary signals and time-varying systems. One windows the signal to extract a relatively small piece of the signal around the time of interest and studies the frequency properties of the extracted piece. If one extracts a very small piece to achieve good time localization the bandwidth of that piece will be artificially large because of the uncertainty principle. Heisenberg uncertainty principle [23] is a classical uncertainty principle, which shows that an increase in the time resolution will result in a decrease in the frequency resolution, and vice versa. In other words, time and frequency resolution can not be maximized at the same time. There could be a reasonable harmonization between time and frequency resolution by means of optimizing the analysis window width adapting to the characteristic of signal. The purpose of this section is to give some new uncertainty principles associated with the STLCT by the precise mathematical formulation.

3.1. Some preparations

We first present the following contents.

Definition 3. Let $f(t)$ be a complex signal, and $E=\int_{-\infty}^{+\infty}|f(t)|^2\mathrm{d}t=\int_{-\infty}^{+\infty}|L_A^f(u)|^2\mathrm{d}u=\int_{-\infty}^{+\infty}|\widehat{f}(\omega)|^2\mathrm{d}\omega$ is the energy of the signal, then we can define [22,33,43]

$$\overline{t_f}=\frac{1}{E}\int_{-\infty}^{+\infty}t\,|\,f(t)\,|^2\,\mathrm{d}t, \tag{18}$$

$$\overline{\omega_f}=\frac{1}{E}\int_{-\infty}^{+\infty}\omega\,|\widehat{f}(\omega)\,|^2\,\mathrm{d}\omega, \tag{19}$$

$$\overline{u_{A,f}}=\frac{1}{E}\int_{-\infty}^{+\infty}u\,|\,L_A^f(u)\,|^2\,\mathrm{d}u. \tag{20}$$

$$T_f^2=\frac{1}{E}\int_{-\infty}^{+\infty}(t-\overline{t_f})^2\,|\,f(t)\,|^2\,\mathrm{d}t, \tag{21}$$

$$F_f^2=\frac{1}{E}\int_{-\infty}^{+\infty}(\omega-\overline{\omega_f})^2\,|\widehat{f}(\omega)\,|^2\,\mathrm{d}\omega, \tag{22}$$

$$F_{A,f}^2=\frac{1}{E}\int_{-\infty}^{+\infty}(u-\overline{u_{A,f}})^2\mid L_A^f(u)\mid^2\mathrm{d}u, \tag{23}$$

where $\widehat{f}(\omega)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}f(t)e^{-it\omega}\mathrm{d}\omega$ is the FT of $f(t)$, T_f^2, F_f^2 and $F_{A,f}^2$ are the spreads of the signal in the time, frequency and LCT domains, respectively. $\overline{t_f}$, $\overline{\omega_f}$ and $\overline{u_{A,f}}$ are the moments (mean) of the signal in the time, frequency and LCT domains, respectively.

Definition 4. Let $f(t)$ be a complex signal and g be a window function then we can define

$$\overline{t_{A,S}}=\frac{1}{\|S_g^Af(t,u)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}t\mid S_g^Af(t,u)\mid^2\mathrm{d}t\mathrm{d}u, \tag{24}$$

$$\overline{u_{A,S}}=\frac{1}{\|S_g^Af(t,u)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}u\mid S_g^Af(t,u)\mid^2\mathrm{d}t\mathrm{d}u, \tag{25}$$

$$T_{A,S}^2=\frac{1}{\|S_g^Af(t,u)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(t-\overline{t_{A,S}})^2\mid S_g^Af(t,u)\mid^2\mathrm{d}t\mathrm{d}u, \tag{26}$$

$$F_{A,S}^2=\frac{1}{\|S_g^Af(t,u)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(u-\overline{u_{A,S}})^2\mid S_g^Af(t,u)\mid^2\mathrm{d}t\mathrm{d}u, \tag{27}$$

where $T_{A,S}^2$ and $F_{A,S}^2$ are the spreads of the signal in the time and STLCT domains, respectively. $\overline{t_{A,S}}$ and $\overline{u_{A,S}}$ are the moments (mean) of the signal in the time and STLCT domains, respectively.

Next, we obtain the following lemma.

Lemma 2. *Let $g\in L^2(\mathbb{R})$ be a window function. For a complex signal $f\in L^2(\mathbb{R})$, we have*

$$\overline{t_{A,S}}=\overline{t_f}+\overline{t_g}, \tag{28}$$

$$T_{A,S}^2=T_f^2+T_g^2, \tag{29}$$

$$\overline{u_{A,S}}=\overline{u_{A,f}}-\overline{u_{A_1,g}}, \tag{30}$$

$$F_{A,S}^2=F_{A,f}^2+F_{A_1,g}^2. \tag{31}$$

For a window function $g(t)$ and its LCT, where $\overline{t_g}$ and $\overline{u_{A_1,g}}$ are the mean time and mean frequency, T_g^2 and $F_{A_1,g}^2$ are the time and frequency spread, respectively. Their definitions are similar to Eq. (18), Eq. (20), Eq. (21) and Eq. (23), but the matrix that corresponds to $g(t)$ is $A_1=(0,b',-\frac{1}{b'},d')$, where $0\neq b'=b\in\mathbb{R}$, the parameter b in the matrix $A=(a,b,c,d)$ and the parameter $d'\in\mathbb{R}$.

Proof. We just derive Eq. (28) and Eq. (31) here, and the rest two can be derived in a similar method.

First, using Lemma 1 and Eq. (4), we have

$$\begin{aligned}\|S_g^Af(t,u)\|^2&=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}|S_g^Af(t,u)|^2\mathrm{d}t\mathrm{d}u\\&=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\Big[\int_{-\infty}^{+\infty}L_A^f(\xi_1)\sqrt{-i2\pi b}e^{i\frac{d'}{2b}(\xi_1-u)^2}\\&\quad\times L_{A_1}^g(\xi_1-u)^*K_A(t,u)K_A^*(t,\xi_1)\mathrm{d}\xi_1\Big]\\&\quad\times\Big[\int_{-\infty}^{+\infty}L_A^f(\xi_2)\sqrt{-i2\pi b}e^{i\frac{d'}{2b}(\xi_2-u)^2}\\&\quad\times L_{A_1}^g(\xi_2-u)^*K_A(t,u)K_A^*(t,\xi_2)\mathrm{d}\xi_2\Big]^*\mathrm{d}t\mathrm{d}u\\&=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}|L_A^f(\xi_1)|^2|L_{A_1}^g(\xi_1-u)|^2\mathrm{d}\xi_1\mathrm{d}u.\end{aligned} \tag{32}$$

Let $\xi_1=\rho$ and $\xi_1-u=\gamma$, then

$$\begin{aligned}\|S_g^Af(t,u)\|^2&=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}|L_A^f(\rho)|^2|L_{A_1}^g(\gamma)|^2\mathrm{d}\rho\mathrm{d}\gamma\\&=\|L_A^f(\rho)\|^2\|L_{A_1}^g(\gamma)\|^2.\end{aligned} \tag{33}$$

Next, by Definition 2 and Eq. (4), we have

$$\begin{aligned}\overline{t_{A,S}}&=\frac{1}{\|S_g^Af(t,u)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}t\mid S_g^Af(t,u)\mid^2\mathrm{d}t\mathrm{d}u\\&=\frac{1}{\|L_A^f(\rho)\|^2\|L_{A_1}^g(\gamma)\|^2}\\&\quad\times\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}t\left[\int_{-\infty}^{+\infty}f(\xi_1)\overline{g(\xi_1-t)}K_A(\xi_1,u)\mathrm{d}\xi_1\right]\\&\quad\times\left[\int_{-\infty}^{+\infty}f(\xi_2)\overline{g(\xi_2-t)}K_A(\xi_2,u)\mathrm{d}\xi_2\right]^*\mathrm{d}t\mathrm{d}u\\&=\frac{1}{\|L_A^f(\rho)\|^2\|L_{A_1}^g(\gamma)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}t|f(\xi_1)|^2|g(\xi_1-t)|^2\mathrm{d}\xi_1\mathrm{d}t.\end{aligned} \tag{34}$$

Let $\xi_1-t=\nu$, we can obtain the mean time of STLCT

$$\begin{aligned}\overline{t_{A,S}}&=\frac{1}{\|L_A^f(\rho)\|^2\|L_{A_1}^g(\gamma)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(\xi_1-\nu)|f(\xi_1)|^2|g(\nu)|^2\mathrm{d}\xi_1\mathrm{d}\nu\\&=\frac{1}{\|L_A^f(\rho)\|^2}\int_{-\infty}^{+\infty}\xi_1|f(\xi_1)|^2\mathrm{d}\xi_1-\frac{1}{\|L_{A_1}^g(\gamma)\|^2}\int_{-\infty}^{+\infty}\nu|g(\nu)|^2\mathrm{d}\nu\\&=\overline{t_f}-\overline{t_g}.\end{aligned} \tag{35}$$

According to the same method, Eq. (30) can be obtained.

Using Eq. (33), Eq. (27) and Eq. (30), the frequency spread of STLCT is shown

$$
\begin{aligned}
F_{A,S}^2 &= \frac{1}{\|S_g^A f(t,u)\|^2}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(u-\overline{u_{A,S}})^2 \mid S_g^A f(t,u)\mid^2 \mathrm{d}t\mathrm{d}u \\
&= \frac{1}{\|L_A^f(\rho)\|^2}\int_{-\infty}^{+\infty}(\rho-\overline{u_{A,f}})^2|L_A^f(\rho)|^2\mathrm{d}\rho + \frac{1}{\|L_{A_1}^g(\gamma)\|^2} \\
&\times \int_{-\infty}^{+\infty}(\gamma-\overline{u_{A_1,g}})|L_{A_1}^g(\gamma)|^2\mathrm{d}\gamma - 2\frac{1}{\|L_A^f(\rho)\|^2} \\
&\times \int_{-\infty}^{+\infty}(\rho-\overline{u_{A,f}})|L_A^f(\rho)|^2\mathrm{d}\rho\frac{1}{\|L_{A_1}^g(\gamma)\|} \\
&\times \int_{-\infty}^{+\infty}(\gamma-\overline{u_{A_1,g}})|L_{A_1}^g(\gamma)|^2\mathrm{d}\gamma \\
&= F_{A,f}^2 + F_{A_1,g}^2 - 2(\overline{u_{A,f}}-\overline{u_{A,f}})(\overline{u_{A_1,g}}-\overline{u_{A_1,g}}) \\
&= F_{A,f}^2 + F_{A_1,g}^2.
\end{aligned} \tag{36}
$$

According to the same method, Eq. (29) can be obtained. □

The Lemma 2 shows that how the duration and bandwidths of the spectrogram are related to the duration and bandwidths of the signal and window.

3.2. Uncertainty principle in time and frequency domains of the STLCT

The uncertainty principles of the LCT are already known. Dang, Deng and Qian [33] generalized the uncertainty principles of the LCT for complex signal in time and frequency domains. It can be restated as follows:

Lemma 3. *Let $f \in L^2(\mathbb{R}), tf(t) \in L^2(\mathbb{R})$ and $A = (a,b,c,d)$ be a matrix parameter satisfying $a,b,c,d \in \mathbb{R}$ and $ad - bc = 1$, then*

$$
T_f^2 F_{A,f}^2 \geq b^2(\frac{1}{4} + COV_f^2 - Cov_f^2) + (aT_f^2 + bCov_f)^2, \tag{37}
$$

where T_f^2 and $F_{A,f}^2$ are defined in Eq. (21) and Eq. (23). Cov_f denotes the covariance of the signal $f(t) = x(t)e^{i\phi(t)}$,

$$
Cov_f = \int_{-\infty}^{+\infty}(t-\overline{t_f})(\phi'(t)-\overline{\omega_f})|x(t)|^2\mathrm{d}t,
$$

and the absolute covariance of the signal $f(t) = x(t)e^{i\phi(t)}$ has the form

$$
COV_f = \int_{-\infty}^{+\infty}|(t-\overline{t_f})(\phi'(t)-\overline{\omega_f})||x(t)|^2\mathrm{d}t.
$$

Based on the Lemma 3, we obtain the following main result.

Theorem 1. *Let $g \in L^2(\mathbb{R})$ be a window function and $tg(t) \in L^2(\mathbb{R})$. $A = (a,b,c,d)$ be a matrix parameter satisfying $a,b,c,d \in \mathbb{R}$ and $ad - bc = 1$. For a complex signal $f \in L^2(\mathbb{R})$ and $tf(t) \in L^2(\mathbb{R})$, then*

$$
T_{A,S}^2 F_{A,S}^2 \geq \Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}}, \tag{38}
$$

where $\Gamma = b^2(\frac{1}{4} + COV_f^2 - Cov_f^2) + (aT_f^2 + bCov_f)^2$, $\Delta = b^2(\frac{1}{4} + COV_g^2)$, Cov_g is the covariance of the signal $g(t) = y(t)e^{i\varphi(t)}$ and COV_g is the absolute covariance of the signal $g(t) = y(t)e^{i\varphi(t)}$.

The equality holds if and only if $f = g$, $a = 0$ and $c = -\frac{1}{b}$, where $f(t)$ has one of the following four Gaussian type functions

$$
f(t) = (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_1]},
$$

$$
f(t) = (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_2]},
$$

$$
f(t) = \begin{cases} (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_3]}, & t \geq \overline{t_f} \\ (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_4]}, & t < \overline{t_f} \end{cases},
$$

$$
f(t) = \begin{cases} (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_5]}, & t \geq \overline{t_f} \\ (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_6]}, & t < \overline{t_f} \end{cases},
$$

where $\gamma, \lambda > 0$ and $\rho_z \in \mathbb{R}, z = 1, 2, \cdots, 6$.

Proof. According to Lemma 2, we have $T_{A,S}^2 F_{A,S}^2 = (T_f^2 + T_g^2)(F_{A,f}^2 + F_{A_1,g}^2) = T_f^2 F_{A,f}^2 + T_g^2 F_{A_1,g}^2 + T_f^2 F_{A_1,g}^2 + T_g^2 F_{A,f}^2$. Using the fact: $n^2 + m^2 \geq 2nm$, for $\forall n, m \in \mathbb{R}$, and Lemma 3, we have

$$
\begin{aligned}
T_{A,S}^2 F_{A,S}^2 &= T_f^2 F_{A,f}^2 + T_g^2 F_{A_1,g}^2 + T_f^2 F_{A_1,g}^2 + T_g^2 F_{A,f}^2 \\
&\geq \Gamma + \Delta + 2\sqrt{T_f^2 F_{A,f}^2 T_g^2 F_{A_1,g}^2} \\
&\geq \Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}},
\end{aligned} \tag{39}
$$

the equality is achieved iff $A = A_1$ and $f = g$ satisfied the results in [31,33]. □

Clearly, Eq. (38) shows that the uncertainty principle in time and frequency domains of the STLCT is connected with the matrix parameters, signal and the window function. We note that the Theorem 1 generalizes other previously developed uncertainty relations. Particularly, when $A = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha)$, we can obtain the uncertainty principle of the short-time fractional Fourier transform (STFRFT) [36] for complex signal in time and frequency domains by Theorem 1, which is given in the following corollary.

Corollary 1. *Let $g \in L^2(\mathbb{R})$ be a window function. For a complex signal $f \in L^2(\mathbb{R})$, then*

$$
T_{\alpha,S}^2 F_{\alpha,S}^2 \geq \Gamma_\alpha + \Delta_\alpha + 2(\Gamma_\alpha\Delta_\alpha)^{\frac{1}{2}}, \tag{40}
$$

where $\Gamma_\alpha = \sin\alpha^2(\frac{1}{4} + COV_f^2 - Cov_f^2) + (\cos\alpha T_f^2 + \sin\alpha Cov_f)^2$, $\Delta_\alpha = \sin\alpha^2(\frac{1}{4} + COV_g^2)$.

Moreover, in the particular cases of an STLCT with parameters $(0, 1, -1, 0)$ implementing a uncertainty principle of the STFT [25,44].

3.3. Uncertainty principle in two STLCT domains

Moreover, Dang, Deng and Qian [33] derived the uncertainty principles for complex signal in two LCT domains. It can be restated as follows:

Lemma 4. *Let $f(t) \in L^2(\mathbb{R})$ and $tf(t) \in L^2(\mathbb{R})$, then*

$$
\begin{aligned}
F_{A,f}^2 F_{B,f}^2 \geq{}& (\frac{1}{4} + COV_f^2 - Cov_f^2)(a_1b_2 - a_2b_1)^2 \\
&+ [a_1a_2T_f^2 + b_1b_2F_f^2 + (a_1b_2 + a_2b_1)Cov_f]^2,
\end{aligned} \tag{41}
$$

where $A = (a_1, b_1, c_1, d_1)$, $B = (a_2, b_2, c_2, d_2)$, $a_1d_1 - b_1c_1 = 1$, $a_2d_2 - b_2c_2 = 1$; $F_{A,f}^2$ and $F_{B,f}^2$ are defined in Eq. (23); T_f^2 and F_f^2 are defined in Eq. (21) and Eq. (22).

In the following, we will derive the uncertainty principle for complex signal in two STLCT domains.

Theorem 2. *Let $g \in L^2(\mathbb{R})$ be a window function and $tg(t) \in L^2(\mathbb{R})$. $A = (a_1, b_1, c_1, d_1)$, $B = (a_2, b_2, c_2, d_2)$ be matrixes parameter satisfying $a_1d_1 - b_1c_1 = 1$, $a_2d_2 - b_2c_2 = 1$ and $A \neq B$. For a complex signal $f \in L^2(\mathbb{R})$ and $tf(t) \in L^2(\mathbb{R})$, then*

$$F^2_{A,S}F^2_{B,S} \geq \Phi + \Psi + 2(\Phi\Psi)^{\frac{1}{2}}, \tag{42}$$

where $\Phi = (\frac{1}{4} + COV_f^2 - Cov_f^2)(a_1b_2 - a_2b_1)^2 + [a_1a_2T_f^2 + b_1b_2F_f^2 + (a_1b_2 + a_2b_1)Cov_f]^2$, $\Psi = (b_1b_2F_g^2)^2$; T_f^2 is defined in Eq. (21); F_f^2 and F_g^2 are defined in Eq. (22).

The equality holds if and only if $f = g$, $a_1 = a_2 = 0$, $c_1 = -\frac{1}{b_1}$ and $c_2 = -\frac{1}{b_2}$, where $f(t)$ has one of the following four Gaussian type functions

$$f(t) = (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2 + \overline{\omega_f}t + \rho_1]},$$

$$f(t) = (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2 + \overline{\omega_f}t + \rho_2]},$$

$$f(t) = \begin{cases} (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2 + \overline{\omega_f}t + \rho_3]}, & t \geq \overline{t_f} \\ (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2 + \overline{\omega_f}t + \rho_4]}, & t < \overline{t_f} \end{cases},$$

$$f(t) = \begin{cases} (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2 + \overline{\omega_f}t + \rho_5]}, & t \geq \overline{t_f} \\ (\frac{\gamma}{\pi})^{\frac{1}{4}} e^{-\frac{\pi}{2}(t-\overline{t_f})^2} e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2 + \overline{\omega_f}t + \rho_6]}, & t < \overline{t_f} \end{cases},$$

where $\gamma, \lambda > 0$ and $\rho_z \in \mathbb{R}$, $z = 1, 2, \cdots, 6$.

Proof. The proof of Theorem 2 is similar to Theorem 1. □

The Eq. (42) shows that the uncertainty principle in two STLCT domains is connected with two matrices parameters, signal and the window function. According to Theorem 2, the uncertainty principle in two STFRFT [36] domains can be obtained when $A = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha)$, $B = (\cos\beta, \sin\beta, -\sin\beta, \cos\beta)$, which is given in the following corollary.

Corollary 2. *Let $g \in L^2(\mathbb{R})$ be a window function. For a complex signal $f \in L^2(\mathbb{R})$, then*

$$F^2_{\alpha,S}F^2_{\beta,S} \geq \Phi_{\alpha,\beta} + \Psi_{\alpha,\beta} + 2(\Phi_{\alpha,\beta}\Psi_{\alpha,\beta})^{\frac{1}{2}}, \tag{43}$$

where $\Phi_{\alpha,\beta} = (\frac{1}{4} + COV_f^2 - Cov_f^2)(\cos\alpha\sin\beta - \cos\beta\sin\alpha)^2 + [\cos\alpha\cos\beta T_f^2 + \sin\alpha\sin\beta F_f^2 + (\cos\alpha\sin\beta + \cos\beta\sin\alpha)Cov_f]^2$ and $\Psi_{\alpha,\beta} = (\sin\alpha\sin\beta F_f^2)^2$.

In addition, in the particular cases of an STLCT with parameters (0,1,-1,0) implementing a uncertainty principle of the STFT [25,44].

3.4. Uncertainty principle for the two conditional standard deviations of the spectrogram

In order to obtain an uncertainty relation for the two conditional standard deviations of the spectrogram, we first give the mean averages and conditional standard deviations, as follows:

$$\langle u \rangle_t = \frac{1}{Q(t)} \int_{-\infty}^{+\infty} u|S_g^A f(t,u)|^2 \mathrm{d}u, \tag{44}$$

$$\langle t \rangle_u = \frac{1}{P(u)} \int_{-\infty}^{+\infty} t|s_g^A f(t,u)|^2 \mathrm{d}t, \tag{45}$$

$$\sigma^2_{u|t} = \frac{1}{Q(t)} \int_{-\infty}^{+\infty} (u - \langle u \rangle_t)^2 |S_g^A f(t,u)|^2 \mathrm{d}u, \tag{46}$$

$$\sigma^2_{t|u} = \frac{1}{P(u)} \int_{-\infty}^{+\infty} (t - \langle t \rangle_u)^2 |s_g^A f(t,u)|^2 \mathrm{d}t. \tag{47}$$

Moreover, let H be a Hilbert space with inner product $\langle\cdot,\cdot\rangle$ given by

$$\langle f, g \rangle = \int_{\mathbb{R}} f(t)g^*(t)\mathrm{d}t.$$

Meanwhile, let $\mathcal{A}$ be a linear operator with domain $D(\mathcal{A}) \subseteq H$ and range in H. The adjoint of the operator $\mathcal{A}$, $\mathcal{A}^*$ can be defined as [45]

$$\langle \mathcal{A}f, g \rangle = \langle f, \mathcal{A}^*g \rangle, \quad \forall f(t) \in D(\mathcal{A}), g(t) \in D(\mathcal{A}^*).$$

$\mathcal{A}$ is said to be Hermitian (or self-adjoint) if $\mathcal{A} = \mathcal{A}^*$.

Theorem 3. *Let $g \in L^2(\mathbb{R})$ be a window function. For a complex signal $f \in L^2(\mathbb{R})$, then*

$$\sigma^2_{u|t}\sigma^2_{t|u} \geq \frac{1}{Q(t)P(u)} \left| \int_{-\infty}^{+\infty} (S_g^A f(t,u))^* (\frac{1}{2}[\mathcal{A},\mathcal{B}]_+ + \frac{i}{2}) f_t(\tau)\mathrm{d}\tau \right|^2, \tag{48}$$

where $\mathcal{A}, \mathcal{B}$ are the Hermitian operators.

Proof. According to Eq. (46), we have

$$\begin{aligned} \sigma^2_{u|t} &= \frac{1}{Q(t)} \int_{-\infty}^{+\infty} (u - \langle u \rangle_t)^2 S_g^A f(t,u)(S_g^A f(t,u))^* \mathrm{d}u \\ &= \frac{1}{Q(t)} \int_{-\infty}^{+\infty} (u - \langle u \rangle_t)^2 \int_{-\infty}^{+\infty} f(\tau)g^*(\tau - t)K_A(\tau,u)\mathrm{d}\tau \\ &\times \int_{-\infty}^{+\infty} f(\tau')g^*(\tau' - t)K_A(\tau',u)\mathrm{d}\tau'\mathrm{d}u \\ &= \frac{1}{Q(t)} \int_{-\infty}^{+\infty} (\frac{b}{i}\frac{d}{d\tau} - \langle u \rangle_t)^2 f(\tau)g^*(\tau - t)(f(\tau)g^*(\tau - t))^*\mathrm{d}\tau \\ &= \frac{1}{Q(t)} \int_{-\infty}^{+\infty} \left|(\frac{b}{i}\frac{d}{d\tau} - \langle u \rangle_t) f_t(\tau)\right|^2 \mathrm{d}\tau. \end{aligned} \tag{49}$$

Let

$$\mathcal{A} = \tau - \langle t \rangle_u; \quad \mathcal{B} = \frac{b}{i}\frac{d}{d\tau} - \langle u \rangle_t, \tag{50}$$

then $\mathcal{A}, \mathcal{B}$ are the Hermitian operators [42,45,46].

So we have

$$\mathcal{AB} = \frac{1}{2}[\mathcal{A},\mathcal{B}]_+ + \frac{i}{2}([\mathcal{A},\mathcal{B}]/i), \tag{51}$$

and

$$[\mathcal{A},\mathcal{B}] = i. \tag{52}$$

By Eq. (47), Eq. (49) and Eq. (50), we obtain

$$\sigma_{t|u}^2\sigma_{u|t}^2=\frac{1}{P(u)}\int_{-\infty}^{+\infty}\mathcal{A}^2|S_g^A f(t,u)|^2\mathrm{d}\tau\frac{1}{Q(t)}\int_{-\infty}^{+\infty}\mathcal{B}^2|f_t(\tau)|^2\mathrm{d}\tau. \tag{53}$$

According to the Schwarz inequality

$$\int_{-\infty}^{+\infty}|f(t)|^2\mathrm{d}t\int_{-\infty}^{+\infty}|g(t)|^2\mathrm{d}t\ge\left|\int_{-\infty}^{+\infty}f^*(t)g(t)\mathrm{d}t\right|^2, \tag{54}$$

we can obtain

$$\begin{aligned}\sigma_{t|u}^2\sigma_{u|t}^2&=\frac{1}{Q(t)P(u)}\int_{-\infty}^{+\infty}\mathcal{A}^2|S_g^A f(t,u)|^2\mathrm{d}\tau\int_{-\infty}^{+\infty}\mathcal{B}^2|f_t(\tau)|^2\mathrm{d}\tau\\&\ge\frac{1}{Q(t)P(u)}\left|\int_{-\infty}^{+\infty}(\mathcal{A}S_g^A f(t,u))^*\mathcal{B}f_t(\tau)\mathrm{d}\tau\right|^2\\&=\frac{1}{Q(t)P(u)}\left|\int_{-\infty}^{+\infty}(S_g^A f(t,u))^*\mathcal{A}\mathcal{B}f_t(\tau)\mathrm{d}\tau\right|^2.\end{aligned} \tag{55}$$

Using Eq. (51) and Eq. (52), we have

$$\begin{aligned}\sigma_{t|u}^2\sigma_{u|t}^2&\ge\frac{1}{Q(t)P(u)}\left|\int_{-\infty}^{+\infty}(S_g^A f(t,u))^*(\frac{1}{2}[\mathcal{A},\mathcal{B}]_+\right.\\&\left.\quad+\frac{i}{2}([\mathcal{A},\mathcal{B}]/i))f_t(\tau)\mathrm{d}\tau\right|^2\\&=\frac{1}{Q(t)P(u)}\left|\int_{-\infty}^{+\infty}(S_g^A f(t,u))^*(\frac{1}{2}[\mathcal{A},\mathcal{B}]_+\right.\\&\left.\quad+\frac{i}{2})f_t(\tau)\mathrm{d}\tau\right|^2.\quad\square\end{aligned} \tag{56}$$

With regard to terminology, sometimes for the sake of emphasis our results, we will call the Theorem 1 and Theorem 2 the global uncertainty principles [26,44,47], and the Theorem 3 the local uncertainty principle [26,44,47].

The novelties of our work are as follows: the results can be viewed as a generalized form of other transforms uncertainty relations, such as the STFRFT (when $A=(\cos\alpha,\sin\alpha,-\sin\alpha,\cos\alpha)$) and STFT (when $A=(0,1,-1,0)$). These novel results show that the uncertainty principles of the STLCT are constrained by the signal, window function and matrixes parameter. However, the situations of the LCT [33] are related to the signal and matrixes parameter. They can be considered as the fact that different constrained factors lead to the different result for the STLCT and the LCT. Thus, they have the different implications. In addition, the work reported in [34,35] were mainly focused on the global uncertainty principles for the STLCT of the real signals. Other kinds of uncertainty principles for the STLCT, such as the local uncertainty principle, have not been reported in the literature. In this paper, we obtain the global uncertainty principles for the STLCT of the complex signals. They are different from the [34,35], our results are related to covariances. However, the concepts of covariance of a signal are important to measure how time and instantaneous frequency are related [41,27]. Moreover, the local uncertainty principle for the STLCT of the complex signals is explored.

The limitations of proposed method: based on the uncertainty principles of the LCT, we obtain the uncertainty principles of the STLCT. Therefore, the Theorem 1 and Theorem 2 depend on the uncertainty principles of the LCT.

3.5. Example

Let a signal is given by

$$f(t)=\begin{cases}(\frac{\gamma}{\pi})^{\frac{1}{4}}e^{-\frac{\pi}{2}(t-\overline{t_f})^2}e^{i[-\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_5]}, & t\ge\overline{t_f}\\(\frac{\gamma}{\pi})^{\frac{1}{4}}e^{-\frac{\pi}{2}(t-\overline{t_f})^2}e^{i[\frac{\lambda}{2}(t-\overline{t_f})^2+\overline{\omega_f}t+\rho_6]}, & t<\overline{t_f}\end{cases},$$

and the window function g is equal to the signal.

According to Definition 3, we have

$$T_f^2=T_g^2=\frac{1}{2\gamma},\quad F_{A,f}^2=\frac{a^2+b^2(\gamma^2+\lambda^2)}{2\gamma},$$

$$F_{A_1,g}^2=\frac{b^2(\gamma^2+\lambda^2)}{2\gamma},$$

$$F_f^2=F_g^2=\frac{\gamma^2+\lambda^2}{2\gamma},\quad Cov_f=0,\quad COV_f=\frac{\lambda}{2\gamma}.$$

Then, we obtain

$$T_{A,S}^2F_{A,S}^2=\frac{1}{\gamma}\left(\frac{a^2+2b^2(\gamma^2+\lambda^2)}{2\gamma}\right)=\frac{a^2}{2\gamma^2}+\frac{b^2(\gamma^2+\lambda^2)}{\gamma^2},$$

$$\Gamma=\frac{b^2(\gamma^2+\lambda^2)}{4\gamma^2}+\frac{a^2}{4\gamma^2},\quad\Delta=\frac{b^2(\gamma^2+\lambda^2)}{4\gamma^2},$$

$$2(\Gamma\Delta)^{\frac{1}{2}}=\frac{1}{2}\left[\left(\frac{b^2(\gamma^2+\lambda^2)}{\gamma^2}\right)^2+\frac{a^2b^2(\gamma^2+\lambda^2)}{\gamma^4}\right]^{\frac{1}{2}}.$$

If $a=0$, then

$$\begin{aligned}\Gamma+\Delta+2(\Gamma\Delta)^{\frac{1}{2}}&=\frac{b^2(\gamma^2+\lambda^2)}{4\gamma^2}+\frac{b^2(\gamma^2+\lambda^2)}{4\gamma^2}+\frac{b^2(\gamma^2+\lambda^2)}{2\gamma^2}\\&=\frac{b^2(\gamma^2+\lambda^2)}{\gamma^2}\\&=T_{A,S}^2F_{A,S}^2.\end{aligned}$$

The above formula indicates the equality in Theorem 1 hold for the signal $f(t)$.

Let $\lambda=\frac{1}{5},\gamma=\frac{1}{2}$, $(a,b,c,d)=(0,\frac{1}{2},-2,1)$ in this example, we can get that

$$T_{A,S}^2F_{A,S}^2=0.29,\quad\Gamma=0.0725,\quad\Delta=0.0725,\quad2(\Gamma\Delta)^{\frac{1}{2}}=0.145,$$

so we can obtain that

$$T_{A,S}^2F_{A,S}^2=\Gamma+\Delta+2(\Gamma\Delta)^{\frac{1}{2}}.$$

If $a\neq0$, then

$$\begin{aligned}M&=T_{A,S}^2F_{A,S}^2-\Gamma-\Delta\\&=\frac{a^2}{2\gamma^2}+\frac{b^2(\gamma^2+\lambda^2)}{\gamma^2}-\left(\frac{b^2(\gamma^2+\lambda^2)}{4\gamma^2}+\frac{a^2}{4\gamma^2}\right)\\&\quad-\left(\frac{b^2(\gamma^2+\lambda^2)}{4\gamma^2}\right)\\&=\frac{a^2}{4\gamma^2}+\frac{b^2(\gamma^2+\lambda^2)}{2\gamma^2}.\end{aligned}$$

$$N=2(\Gamma\Delta)^{\frac{1}{2}}=\frac{1}{2}\left[\left(\frac{b^2(\gamma^2+\lambda^2)}{\gamma^2}\right)^2+\frac{a^2b^2(\gamma^2+\lambda^2)}{\gamma^4}\right]^{\frac{1}{2}}.$$

Let $m=\frac{b^2(\gamma^2+\lambda^2)}{\gamma^2}$, then

$$M=\frac{a^2}{4\gamma^2}+\frac{m}{2}>0,\quad N=\frac{1}{2}\left[m^2+\frac{a^2m}{\gamma^2}\right]^{\frac{1}{2}}>0.$$

Hence

$$M^2 - N^2 = \frac{a^4}{16\gamma^4} > 0,$$

we have $M > N$. So

$$T^2_{A,S}F^2_{A,S} > \Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}}$$

The above formula indicates the inequality in Theorem 1 hold for the signal $f(t)$.

Let $\lambda = \gamma = 1$, $(a, b, c, d) = (2, \frac{1}{4}, -1, 1)$ in this example, we can get that

$$T^2_{A,S}F^2_{A,S} = 2.125, \quad \Gamma = 1.03125, \quad \Delta = 0.03125,$$

$$2(\Gamma\Delta)^{\frac{1}{2}} = 0.35903516,$$

so we can obtain that

$$T^2_{A,S}F^2_{A,S} > \Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}}.$$

Moreover

$$\begin{aligned} F^2_{A,S}F^2_{B,S} &= \frac{a_1^2 + 2b_1^2(\gamma^2+\lambda^2)}{2\gamma} \times \frac{a_2^2 + 2b_2^2(\gamma^2+\lambda^2)}{2\gamma} \\ &= \frac{a_1^2a_2^2 + 2a_1^2b_2^2V + 2a_2^2b_1^2V + 4b_1^2b_2^2V^2}{4\gamma^2}, \end{aligned}$$

$$\begin{aligned} \Phi &= \frac{V}{4\gamma^2}(a_1b_2 - a_2b_1)^2 + \left(\frac{a_1a_2}{2\gamma} + \frac{b_1b_2V}{2\gamma}\right)^2 \\ &= \frac{V}{4\gamma^2}a_1^2b_2^2 + \frac{V}{4\gamma^2}a_2^2b_1^2 + \frac{a_1^2a_2^2}{4\gamma^2} + \frac{V^2}{4\gamma^2}b_1^2b_2^2, \end{aligned}$$

and

$$\Psi = \frac{V^2}{4\gamma^2}b_1^2b_2^2,$$

where $V = \gamma^2 + \lambda^2$.

Then

$$Y = F^2_{A,S}F^2_{B,S} - \Phi - \Psi = \frac{V}{4\gamma^2}a_1^2b_2^2 + \frac{V}{4\gamma^2}a_2^2b_1^2 + \frac{V^2}{2\gamma^2}b_1^2b_2^2 > 0,$$

and

$$\begin{aligned} H^2 &= \left[2(\Phi\Psi)^{\frac{1}{2}}\right]^2 \\ &= \frac{V^3}{4\gamma^4}a_1^2b_1^2b_2^4 + \frac{V^3}{4\gamma^4}a_2^2b_1^4b_2^2 + \frac{V^2}{4\gamma^4}a_1^2a_2^2b_1^2b_2^2 + \frac{V^4}{4\gamma^4}b_1^4b_2^4, \end{aligned}$$

where $H = 2(\Phi\Psi)^{\frac{1}{2}} > 0$.

So

$$\begin{aligned} Y^2 - H^2 &= \frac{V^2}{16\gamma^4}a_1^4b_2^4 + \frac{V^2}{16\gamma^4}a_2^4b_1^4 - \frac{V^2}{8\gamma^4}a_1^2a_2^2b_1^2b_2^2 \\ &= \frac{V^2}{16\gamma^4}(a_1^2b_2^2 - a_2^2b_1^2) \geq 0, \end{aligned}$$

we obtain $Y \geq H$. Hence

$$F^2_{A,S}F^2_{B,S} \geq \Phi + \Psi + 2(\Phi\Psi)^{\frac{1}{2}}.$$

If $a_1 = a_2 = 0$, then

$$F^2_{A,S}F^2_{B,S} = \Phi + \Psi + 2(\Phi\Psi)^{\frac{1}{2}}.$$

The above formula indicates the equality in Theorem 2 hold for the signal $f(t)$.

Let $\lambda = \gamma = 1$, $(a_1, b_1, c_1, d_1) = (0, 2, -\frac{1}{2}, 1)$ and $(a_2, b_2, c_2, d_2) = (0, \frac{1}{5}, -5, 1)$ in this example, we can get that

$$F^2_{A,S}F^2_{B,S} = 0.64, \quad \Phi = 0.16, \quad \Psi = 0.16, \quad 2(\Phi\Psi)^{\frac{1}{2}} = 0.32,$$

so we can obtain that

$$F^2_{A,S}F^2_{B,S} = \Phi + \Psi + 2(\Phi\Psi)^{\frac{1}{2}}.$$

Otherwise, we have

$$F^2_{A,S}F^2_{B,S} > \Phi + \Psi + 2(\Phi\Psi)^{\frac{1}{2}}.$$

It indicates the inequality in Theorem 2 hold for the signal $f(t)$.

Let $\lambda = \gamma = 1$, $(a_1, b_1, c_1, d_1) = (1, 2, -1, 1)$ and $(a_2, b_2, c_2, d_2) = (2, \frac{1}{2}, -1, 2)$ in this example, we can get that

$$F^2_{A,S}F^2_{B,S} = 6.25, \quad \Phi = 1.25, \quad \Psi = 0.25, \quad 2(\Phi\Psi)^{\frac{1}{2}} \approx 1.11803398,$$

so we can obtain that

$$F^2_{A,S}F^2_{B,S} > \Phi + \Psi + 2(\Phi\Psi)^{\frac{1}{2}}.$$

4. Potential applications

The philosophy of the STLCT uncertainty principles is the same as that for the uncertainty principles in the LCT setting. The STLCT case is subject to suitably incorporate window function. In this section, some potential applications are presented to show the useful and importance of the theorems in signal processing.

Uncertainty principles are often used to estimate the bandwidths [22,25,32]. For example, if $T_{A,S}$ is known, then

$$F^2_{A,S} \geq \frac{\Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}}}{T^2_{A,S}} \geq \frac{b^2}{4T^2_{A,S}}. \tag{57}$$

Even the middle term of the chain of the inequality usually cannot be reached, except it is one of four Gaussian type functions given in Theorem 1.

Alternative to Theorem 1, we have

$$\begin{aligned} &\Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}} \\ &= \min\left\{T^2_{A,S}F^2_{A,S} : \int_{\mathbb{R}} |f(t)|^2 dt \right. \\ &\left. = \int_{\mathbb{R}} |g(t)|^2 dt = 1, tf(t) \in L^2(\mathbb{R}), tg(t) \in L^2(\mathbb{R})\right\}. \end{aligned} \tag{58}$$

The relation shows that the minimum value $\Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}}$ of the uncertainty product $T^2_{A,S}F^2_{A,S}$ can be reached. For a practical signal this minimum value should not be reached, and the corresponding uncertainty product is actually larger than $\Gamma + \Delta + 2(\Gamma\Delta)^{\frac{1}{2}}$. If we are dealing with a class of signals satisfying: $T_f \geq \eta > 0$ and $(a, b, c, d) = (1, 0, 1, 1)$, then the uncertainty product cannot be less than η^4.

Uncertainty principles of the STLCT can be used for the analysis of optical systems related to the spread in the STLCT domain. Below we give an example to illustrate the application aspect of the result of uncertainty principle in the STLCT domains. The wave propagation through an aperture, the free-space propagation, and the pulse propagation in optical fibers [22]. We will consider the Theorem 1 in the realm of wave propagation through an aperture. Immediately after crossing the aperture the field has some effective width $T^2_{A,S}$ dictated by the aperture width. After propagation a distance ξ the transversal distribution of the field can be described by a STLCT with parameters matrix $A=(a,b,c,d)=(1,\frac{\epsilon\xi}{2\pi},0,1)$. Then the relation Eq. (37) becomes

$$\begin{aligned}F_{A,S}^2 \geq& \frac{\epsilon^2\xi^2}{4\pi^2 T_{A,S}^2}\left(\frac{1}{4}+COV_f^2-Cov_f^2\right)\\&+\frac{1}{T_{A,S}^2}\left(T_f^2+\frac{\epsilon\xi}{2\pi}Cov_f\right)^2+\frac{\epsilon^2\xi^2}{4\pi^2T_{A,S}^2}\left(\frac{1}{4}+COV_g^2\right)\\&+\frac{\epsilon\xi}{\pi T_{A,S}^2}\Big[\frac{\epsilon^2\xi^2}{4\pi^2}\left(\frac{1}{4}+COV_f^2-Cov_f^2\right)\\&\times\left(\frac{1}{4}+COV_g^2\right)+\left(T_f^2+\frac{\epsilon\xi}{2\pi}Cov_f\right)^2\left(\frac{1}{4}+COV_g^2\right)\Big]^{\frac{1}{2}}\end{aligned}\tag{59}$$

implying that for short ξ the effective spread $F_{A,S}^2$ is slightly larger than that at the aperture plane $T_{A,S}^2$ and for large propagation distances ξ the spread of the field is proportional to $\epsilon\xi$ and reciprocally proportional to the field spread in the aperture plane.

In many applications we are interested in an output with very compact support such as pulse compression, beam focusing, and remote imaging. By Theorem 1 we see that a necessary condition for narrow output is for parameter b to be sufficiently small. For example, in the limiting case $b\to 0$ the STLCT approached a kind of scaling and chirp multiplication operations [40] that simply modulates and magnifies or demagnifies the input function. An ideal imaging system is an example of a system that performs STLCT with $b=0$.

Other potential applications can be found in the estimation of the lower bound of integral. Such as if a signal in Theorem 1 is determined, it is difficult to obtain the value of the left side of Theorem 1 by calculating the integral. However, an estimation of the left side of Theorem 1 can be easily obtained based on Theorem 1.

The uncertainty principles have some applications in signal recovery. Donoho and Stark [28] studied the problem of signal recovery by a uncertainty principle. In order to illustrate the application of uncertainty principle, we first present the uncertainty principle [28]: If a signal f is practically zero outside a measurable set T and its Fourier transform $\widehat{f}(w)=\int_{-\infty}^{+\infty}f(t)e^{-2\pi itw}dt$ is practically zero outside a measurable set W, then

$$|T||W|\geq 1-\delta,\tag{60}$$

where $|T|$ and $|W|$ denote the measures of the sets T and W, respectively, and δ is a small number bound.

According to the Eq. (60), the [28] show that the recovery of a signal despite significant amounts of missing information. One example is: A signal $f\in L^2(\mathbb{R})$ is transmitted to a receiver who knows that $f(t)$ is band-limited, meaning that f was synthesized using only frequencies in a set W. Now suppose the receiver is unable to observe all of s, a certain subset T of t-values is unobserved. Furthermore, the observed signal is contaminated by observational noise $n(t)\in L^2(\mathbb{R})$. Thus, the received signal $r(t)$ satisfies

$$r(t)=\begin{cases}f(t)+n(t), & t\notin T\\ 0, & t\in T\end{cases}\tag{61}$$

The receiver's aim is to reconstruct the transmitted signal $f(t)$ from the noisy received signal $r(t)$. Although it may seem that information about $f(t)$ for $t\in T$ is unavailable, the uncertainty principles say that recovery is possible provided $|T||W|\geq 1$. Donoho and Stark [28] proved this result in the Fourier transform case, and Liu et al. [44] generalized the result to the STFT. We may derive the analogue result to STLCT. Hence, the application of the uncertainty principles for STLCT in signal recovery is our main research direction in the future.

In addition, the STLCT is an affine transformation in the time-frequency plane. The uncertainty principles of the STLCT play a very important role in the analysis of spectra in affine modulation systems [11,39]. For example, when we try to implement the STLCT in affine modulation schemes where the spectral efficiency is crucial with ability to compress pulses within the time-frequency plane, it is necessary to discuss the effective bandwidth in the STLCT domain. It is theoretically meaningful and practically feasible to discuss the spread in the STLCT domain through combining the new uncertainty principle inequalities with the multi-channel interpolation techniques. Moreover, the potential applications also can be found in signal modulations and filter design, such as the methods presented in [24,48,49].

In this paper, in order to obtain the uncertainty principles of the STLCT, we first need the Lemma 2. The method of the Lemma 2 follows the following three steps:

(i) According to the Definition 1 and the Definition 2, the relationships between the STLCT of the signal f and the LCTs of the signal f and the window function g can be obtained i.e. Eq. (15).

(ii) Based on the relationships and the definition of the normal, we have the normal of the STLCT i.e. Eq. (33).

(iii) From the Definition 3 and Definition 4, we obtain the relationships i.e. Lemma 2.

Then, based on the Lemma 2, we obtain the Theorem 1 and Theorem 2. The detail algorithm of the method as follows: The product of $T_{A,S}^2$ (the spread of the signal in the time domain) and $F_{A,S}^2$ (the spread of the signal in the STLCT domain) can be computed via the Lemma 2. Applying the fact that $n^2+m^2\geq 2nm$, for $\forall n,m\in\mathbb{R}$ and Lemma 3, it follows that Eq. (38) can be obtained. When the signal f is Gaussian type function, the equality holds in Lemma 3. In addition, from $n^2+m^2\geq 2nm$, the equality holds if and only if $m=n$. Hence, we obtain the conditions of the equality holds in Theorem 1. Similarly, the product of $F_{A,S}^2$ and $F_{B,S}^2$ (the spreads of the signal in two STLCT domains) can be computed via the Lemma 2. Applying the fact that $n^2+m^2\geq 2nm$, for $\forall n,m\in\mathbb{R}$ and Lemma 4, then Eq. (42) can be obtained. When the signal f is Gaussian type function, the equality holds in Lemma 4. Moreover, we say that the equality holds of $n^2+m^2\geq 2nm$ if $m=n$. Hence, we obtain the conditions of the equality holds in Theorem 2.

Furthermore, based on Eq. (46) and the definition of the STLCT, Eq. (49) can be obtained. Applying the properties of the Hermitian operators and Eq. (47), we have the product of the two conditional standard deviations of the spectrogram i.e. Eq. (53). Eq. (53) can be computed by Schwarz inequality. Finally, in accordance with the properties of the Hermitian operators, Theorem 3 can be given.

It can provide convenience for studying the uncertainty principles of the other form transforms, such as Wigner distribution (WD) and ambiguity function (AF) [50,51].

5. Conclusions

In this paper, many different forms of uncertainty principles for complex signals associated with the STLCT are derived. Firstly, based on the relation between the STLCT and the LCT, a new uncertainty principle for the STLCT of complex signals in time and frequency domains is derived. Secondly, the uncertainty principle in two STLCT domains is obtained. We formulate these uncertainty principles are related to the covariance of time and frequency and can be achieved by complex chirp signals with Gaussian signals. In addition, according to the Hermitian operators, the uncertainty principle for the two conditional standard deviations of the spectrogram associated with the STLCT is discussed. Example is given to illustrate our results. Finally, some potential applications of the derived theorems are proposed. These uncertainty principles are new results. Our further work will think about these applications of the STLCT in signal processing and these uncertainty principles for discrete signals. Moreover, we will strive to study the reasons for the existence of different bounds for Heisenberg's uncertainty principles for real and complex signals.

CRediT authorship contribution statement

Wen-Biao Gao: Conceptualization, Formal analysis, Methodology, Validation, Writing – original draft. **Bing-Zhao Li:** Funding acquisition, Project administration, Resources, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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